ARMORER'S BENCH TECHNICAL MANUAL

Lee-Enfield No. 4 Mk I & SMLE Mk III

RSAF Enfield / BSA / Long Branch / Savage-Stevens • British Commonwealth Rapid-Fire Battle Rifle (.303 British)

Manufacturer:	RSAF Enfield / BSA / Long Branch / Savage-Stevens	**Category:**	Rifles & Carbines
Action Type:	Cock-on-Closing Rear-Locking Bolt Action	**Era / Service:**	British Empire Service (1895–1957+)
Calibers:	.303 British (Mk VII Rimmed)	**Covered Models:**	SMLE No. 1 Mk III* (WWI), Rifle No. 4 Mk I (WWII), No. 4 Mk 2, No. 5 Mk I 'Jungle Carbine'

KEY ARMORER TASKS & BENCH OPERATIONS

Core Procedures: Numbered Bolt Head Swapping (0, 1, 2, 3 Headspace) • Striker Collar & Firing Pin Removal Tool • Forend Stock Draw Block Bedding • Magazine Feed Lip Bending Tool • Two-Stage Military Trigger Tuning

CRITICAL ARMORER TOLERANCES & SPECIFICATIONS

SPECIFICATION / CLEARANCE	ARMORER TOLERANCE & BENCH SPECIFICATION
HEADSPACE ADJUSTMENT	Set by unscrewing and swapping numbered bolt heads: Size 0 (standard), Size 1 (+0.003"), Size 2 (+0.006"), Size 3 (+0.009")
HEADSPACE LIMITS	GO = 0.058" / NO-GO = 0.064" / FIELD (Military Service Limit) = 0.070"
COCK ON CLOSING	Bolt cocks on the forward stroke (allowing the famous 'Mad Minute' 30+ aimed shots in 60 seconds)
REAR LOCKING LUGS	Dual locking lugs at rear of bolt body; produces natural micro-flex in receiver, requiring exact forend draw block bedding

IMMEDIATE ACTION & MALFUNCTION DIAGNOSTIC MATRIX

MALFUNCTION CONDITION	PROBABLE CAUSES	ARMORER BENCH REMEDY
Stoppage 1: Excessive Headspace / Primer Back-Out on .303 **Fired casings show stretched case head web or primer backing out of pocket by 0.010"**	• Wear on rear receiver locking shoulders from decades of firing high-pressure military cordite ammunition. • Worn Size 0 bolt head.	**Gunsmith Action: Measure headspace with 0.064" NO-GO gauge. If bolt closes on NO-GO, unscrew bolt head from bolt body. Replace Size 0 bolt head with Size 1 or Size 2 bolt head. Re-gauge until bolt resists NO-GO gauge.**

CURATED BENCH VIDEO REFERENCES

PROCEDURE TITLE	INSTRUCTOR / CREATOR	TECHNICAL PROCEDURE SUMMARY
Lee-Enfield No. 4 Mk I Headspace, Bolt Heads & Complete Teardown	Mark Novak / Anvil Gunsmithing	Mark Novak explains the unique Lee-Enfield numbered bolt head system, measuring rimmed headspace, disassembly of the striker and spring, and bedding draw blocks.